source ..... noise engineering manual
rev 2026.10

sinc legio

_travelers guide
_for syncing the satellites

2	3	6
switches · 3 ways	cv inputs	hp

[pitch] press [pitch] hold › then noon = 12 o'clock tags = the panel drawing

multimode stereo pm oscillator · wavemorph, operator and infinifolder modes · hard sync

← back to the legio starsystem

= hold

01 what it is

si_01

pm osc	stereo phase-mod oscillator, sinc iter reborn
three algos	warp, op and fold
sync	hard sync input
in L R	pm · audio in · L feeds R if alone
out L R	stereo out, up to 16 V peak to peak
platform	a legio firmware · other firmwares

02 pitch

si_02 · enc

[enc] turn	fine tune
[enc] press + turn	coarse tune
enc jack	pitch jack · 1 V/oct · –2 to +5 V

10 first flight

si_10 · the manual's tutorial

1. out to the mixer
2. a pitch sequence into enc
3. s1 warp · s2 0
4. move 1 and 2
5. try op, then fold with chorus

03 morph · timbre

si_03 · knobs 1 + 2

1 [morph]	carrier waveform
2 [timbre · warp]	pitch of an internal modulator, summed in
2 [timbre · op]	pm index · self-pm with nothing in pm
2 [timbre · fold]	fold amount

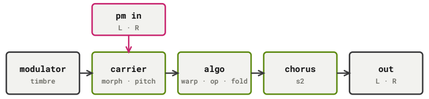
04 algo · chorus

si_04 · s1 · s2 · top → bottom

s1 warp	pm with the modulator mixed in · am-ish
s1 op	a pm operator for chains
s1 fold	infinifolder
s2 chorus	II lots · I some · 0 off

__fig.01 : signal path

in → out



as the manual describes it

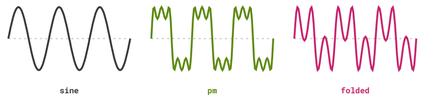
05 jacks

si_05 · trig · cv · audio

trig sync	hard sync · near +1.6 V
in pm L R	audio-rate pm · L feeds R
out L R	up to 16 V peak to peak
cv	pitch –2 to +5 V · morph + timbre 0 – 5 V

__fig.02 : waveforms

shape · mod



08 patch ideas

si_08 · the manual

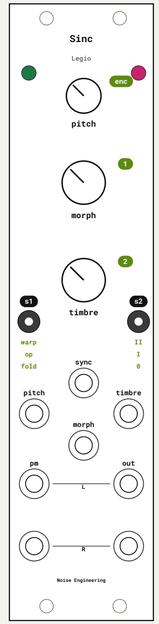
bubbly saws — warp · 0
morph and timbre to taste

multi-op fm — op · 0
two or more legios in series · timbre = pm amount

super stereo — fold · I / II
wide folded leads

__fig.03 : the panel

numbers match the legend



- enc **pitch** · fine tune
- 1 **morph** · carrier waveform
- 2 **timbre** · pitch of an internal modulator, su…
- s1 **algo** · warp · op · fold
- s2 **chorus** · II · I · 0
- trig **sync** · hard sync
- cv **pitch** · **1** · **2** · pitch –2…+5 V · others …
- in **pm L / R** · L feeds R if alone
- out **out L / R** · stereo out

09 specs

si_09

size	6 hp · 1.5 in deep
power	140 mA +12 V · 22 mA –12 V
cv in	0 – 5 V · pitch –2 to +5 V
trig	rising edge near +1.6 V
boot LEDs	green / magenta
calibrate	power on with nothing in pitch cv to recalibrate

design homage to the house of x / powers of x data pages — designed by tom muller with jonathan hickman · panel drawing after the noise engineering legio overlay template
not affiliated with noise engineering or marvel

conceived by mellotom · legio-sinc-navigator ... 00.10_01
created with claude